GSD RESEARCH MISSION PACKAGE

The AM Radio Dial
A Broadcast Heritage Study

West Coast · Pacific Northwest · California
History, Licensing, Spectrum & Future

A Deep Research Study

Vision → Research → Mission
Full Pipeline Execution

Date: March 27, 2026 | Status: Ready for GSD Orchestrator
Activation Profile: Fleet | Estimated: 18–24 context windows across 4–6 sessions

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STAGE 1 — VISION DOCUMENT

1 AM Radio: West Coast & PNW — Vision Guide

Date: March 27, 2026

Status: Initial Vision / Pre-Research Phase

Depends On: None (foundational research mission)

Context: Fox Infrastructure Group / GSD Ecosystem Research Pipeline

1.1 Vision

Before the internet, before satellite, before the smartphone—there was a voice in the darkness, carrying itself over hundreds of miles of ocean, mountain, and forest on invisible waves. AM radio was not merely a technology; it was the central nervous system of the American West. It connected loggers in the Cascades to Portland markets, fishermen off the Washington coast to Seattle weather reports, and ranch hands in the Central Valley to the baseball games of the Giants. The Pacific Northwest and California were, from the earliest days of licensed broadcasting, a laboratory for the entire medium.

This research mission traces that arc from the first licensed experimental transmissions in 1920 San Jose to the all-digital AM frontier of 2026. It maps the ecosystem of the AM dial as it developed along the West Coast—the network affiliations, the format wars, the clear-channel titans and the daytimer underdogs—with particular focus on the stations of Washington, Oregon, and California that shaped an entire region's sonic identity. The same curiosity that asks "what was broadcasting before" also asks "what is broadcasting becoming," and this mission addresses both with equal rigor.

The practical thread runs alongside the historical: the mechanics of obtaining a broadcast license today, the open spectrum that remains available, the rules and regulations under which all broadcasters must operate, and the pathways—full-power AM, full-power FM, Low Power FM, FM translators, and Part 15 unlicensed devices—that a new voice can use to reach the air in 2026. The AM dial is not a closed museum; it is a living infrastructure with legal on-ramps still accessible to those who understand the system.

For the builder who hears frequencies as possibility, this document is a flight manual.

1.2 Problem Statement

1. **Fragmented institutional memory.** The history of West Coast AM radio exists in scattered archives, fan sites, local historical societies, and FCC dockets. No single authoritative synthesis covers the full arc from experimental licenses in 1920 to the digital transition of the 2020s with geographic specificity to Washington and California.

2. **Licensing opacity for prospective broadcasters.** The FCC licensing process

for AM, FM, LPFM, and FM translator stations involves multiple forms, filing windows, engineering studies, and regulatory conditions that are individually documented but difficult to navigate as an integrated pathway. The spectrum scarcity situation, particularly in urban West Coast markets, is poorly understood by most non-industry applicants.

3. **Unclear open-spectrum inventory.** As of 2026, the actual availability of AM frequencies in specific PNW and California markets—accounting for Class A clear-channel protections, daytimer restrictions, and NARBA international agreements—is unknown to most prospective licensees and not synthesized in accessible form.
4. **Regulatory landscape complexity.** Broadcast regulations under 47 CFR Part 73 span content obligations (obscenity, indecency, EAS), technical standards (power limits, antenna systems, modulation), ownership rules (local market caps, cross-ownership), and public interest requirements—a body of law that has changed substantially since the Telecommunications Act of 1996 and continues to evolve under AM Revitalization proceedings.
5. **Uncertainty about AM's future viability.** The EV interference problem, declining automotive AM reception, streaming competition, and the HD Radio digital transition create genuine uncertainty about the investment value of AM broadcasting infrastructure in 2026 and beyond—uncertainty that requires evidence-based analysis, not speculation.

1.3 Core Concept

The Broadcast Archaeology Model: Treat the AM dial as a stratigraphic record—each layer of regulation, technology, and format revealing the social and technical forces that shaped it. Map both the deep past and the navigable present on the same coordinate system.

1.3.1 West Coast Broadcast Geography

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WEST COAST AM BROADCAST GEOGRAPHY
    ═══════════════════════════════════════════════════════════


  PACIFIC NORTHWEST ZONE                 CALIFORNIA ZONE
  ┌──────────────────────┐               ┌──────────────────────────┐
  │  SEATTLE             │               │  SAN FRANCISCO BAY       │
  │  KJR (950) est.1921  │               │  KGO/KSFO (810) 1924     │
  │  KNWN/KOMO (1000)    │               │  KNBR (680) 1922         │
  │  KIRO (710)          │               │  KYA (1260) 1926         │
  │  KVI (570)           │               │  KCBS (740) 1909*        │
  │  KLAY/KOL            │               │                          │
  │                      │               │  LOS ANGELES             │
  │  TACOMA              │               │  KFI (640) 1922          │
  │  KMO  KTCL           │               │  KNX (1070) 1920         │
  │                      │               │  KABC (790) 1925         │
```

```
│  PORTLAND              │              │  KMPC (710) 1922       │
│  KGW (620) 1922        │              │                        │
│  KOIN (940)            │              │  SAN DIEGO             │
│  KEX (1190)            │              │  KFSD/KOGO (600)       │
│                        │              │  KCBQ (1170)           │
│  SPOKANE               │              └────────────────────────┘
│  KHQ  KGA  KXLY        │
└────────────────────────┘

FREQUENCY CLASSES (FCC AM)
Class A (Clear Channel) — 50 kW, nationwide nighttime reach
Class B (Regional)      — Up to 50 kW, regional protection
Class C (Local)         — Up to 1 kW
Class D (Daytimer)      — Daytime only, sunset-to-sunrise silent

* KCBS traces origins to experimental station FN in 1909
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1.3.2 Six-Module Research Architecture

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RESEARCH MODULE ARCHITECTURE
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M1: AM BAND          M2: PNW HISTORY       M3: CALIFORNIA
FOUNDATIONS     ──►  STATIONS         ──►  STATIONS
(Regulation,         (1920–spresent,        (1920–spresent,
Technology,          WA/OR/ID focus)        Bay Area + LA)
Band Structure)
     │                    │                      │
     ▼                    ▼                      ▼
M4: SPECTRUM &      M5: FCC RULES &       M6: DIGITAL
LICENSING       ──► REGULATIONS       ──► FUTURE
PATHWAYS             (47 CFR Pt 73,         (HD Radio,
(AM/FM/LPFM,         Ownership, EAS,        AM Revival,
Translators)         Content)               Streaming)

DEPENDENCY CHAIN: M1 → M2 → M3 → M4 → M5 → M6
PARALLEL TRACKS:  M2 ‖ M3,  M4 ‖ M5
```

1.4 Research Modules

1.4.1 Module 1 — AM Band Foundations

The physics of amplitude modulation, the history of the AM band (535–1705 kHz), the Federal Radio Commission of 1927 and its successor the FCC, the Radio Act of 1927, the Communications Act of 1934, the NARBA international frequency assignment treaty, and the AM station classification system (Class A through D). This module establishes the technical and regulatory substrate on which all other modules rest.

1.4.2 Module 2 — Pacific Northwest Historical Stations

A comprehensive station-by-station chronicle of AM broadcasting in Washington, Oregon, and Idaho from the earliest experimental licenses (KJR 1921, KGW Portland 1922) through the Golden Age of network radio, the format revolution of Top 40, the news-talk pivot, and the consolidation era following the Telecommunications Act of 1996. Key stations: KJR, KOMO, KIRO, KVI, KGW, KOIN, KEX, KHQ, KGA, KOL, and regional stations in Tacoma, Bellingham, Everett, Spokane, and Eugene.

1.4.3 Module 3 — California Historical Stations

The parallel chronicle for California, with emphasis on the Bay Area (KGO/KSFO, KNBR, KYA, KCBS) and Los Angeles (KFI, KNX, KABC, KMPC, KFWB). The Don Lee Network, the NBC Pacific Coast network, the NBC Orange Chain, and their West Coast infrastructure. California-specific AM demographics: as of the AM Revitalization NPRM, 34% of California AM stations serve specific minority communities, including 33 Spanish-language, 13 Asian-format, and 4 Korean-format outlets.

1.4.4 Module 4 — Spectrum Availability and Licensing Pathways

The practical guide to entering broadcasting in 2026. Available pathways: (1) Full-power AM station—Class A through D, minimum 250 W, requires construction permit; (2) Full-power FM station—minimum 100 W, requires open allotment and engineering study; (3) Low Power FM (LPFM)—up to 100 W, nonprofit/educational only, 3.5-mile radius, filed during FCC windows; (4) FM Translator—rebroadcasts AM or other content on FM; (5) Part 15 unlicensed—200-foot radius only. Current spectrum scarcity in Seattle, Portland, San Francisco, and Los Angeles makes new full-power stations difficult but not impossible.

1.4.5 Module 5 — FCC Rules, Regulations, and Broadcaster Obligations

The regulatory framework governing all licensees under 47 CFR Part 73. Topics: public interest obligations, the FCC's eight-year license term, Equal Employment Opportunity requirements, political broadcasting obligations (equal time, lowest unit rate), Emergency Alert System (EAS) participation, indecency and obscenity restrictions under 18 U.S.C. § 1464, ownership caps (Telecommunications Act of 1996), construction permit requirements, and operational logs/record-keeping. Recent 2025 FCC rulemaking updating Part 73 references and LPFM separation requirements.

1.4.6 Module 6 — The Digital Future of AM

The trajectory of AM broadcasting: HD Radio (IBOC hybrid digital), the FCC's 2020 approval of all-digital AM in the MA3 mode (FCC 20-154), the AM Revitalization proceeding (Docket 13-249), FM translator access granted to AM stations, the EV radio interference controversy (2023–2024), and the competitive landscape with streaming audio (AM/FM share fell to 36% of audio listening by 2023 per Edison Research). The 95 million vehicles already equipped with HD Radio technology as a counterweight.

1.5 Chipset Configuration

```
chipset:
  name: am_radio_west_coast_fleet
  version: "1.0.0"
  profile: fleet
  modules:
    - id: M1
      name: am_band_foundations
      model: sonnet
      parallel_group: null
      depends_on: []
    - id: M2
      name: pnw_historical_stations
      model: sonnet
      parallel_group: alpha
      depends_on: [M1]
    - id: M3
      name: california_historical_stations
      model: sonnet
      parallel_group: alpha
      depends_on: [M1]
    - id: M4
      name: spectrum_and_licensing
      model: sonnet
      parallel_group: beta
      depends_on: [M1]
    - id: M5
      name: fcc_rules_and_regulations
      model: sonnet
      parallel_group: beta
      depends_on: [M1]
    - id: M6
      name: digital_future_am
      model: opus
      parallel_group: null
      depends_on: [M2, M3, M4, M5]
  synthesis:
    model: opus
    depends_on: [M2, M3, M4, M5, M6]
  publication:
    model: sonnet
    depends_on: [synthesis]
  token_budget_k: 280
  parallel_efficiency: 0.72
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\subsection{Scope Boundaries}

\subsubsection{In Scope (v1.0)}
\begin{itemize}[leftmargin=*]
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```
  \item AM band history on the West Coast (Washington, Oregon, Idaho,
    California) from 1920 to 2026
  \item Station histories for all Class~A and major Class~B AM stations in
     the PNW and California
  \item Complete FCC licensing pathways for AM, FM, LPFM, and FM
    translator stations
  \item Spectrum availability analysis for West Coast markets with
    emphasis on Seattle, Portland, San Francisco, and Los Angeles metro
    areas
  \item 47 CFR Part~73 rules applicable to new and existing broadcast
    licensees
  \item AM digital transition (HD Radio, all-digital MA3) and future
    viability assessment
  \item Open band frequencies and regulatory context for prospective
    broadcasters
\end{itemize}

\subsubsection{Out of Scope}
\begin{itemize}[leftmargin=*]
  \item FM station history (except where directly relevant to AM simulcast
     or translator context)
  \item Television broadcasting history
  \item Stations east of the Rockies except as context for clear-channel
    frequency assignments
  \item Amateur (ham) radio licensing and operations
  \item International broadcasting or shortwave operations
  \item Satellite radio (SiriusXM) except as competitive context
  \item Podcast and streaming platform analysis beyond AM competitive
    landscape framing
\end{itemize}

\subsection{Success Criteria}

\begin{enumerate}[leftmargin=*, label=\textbf{SC-\arabic{enumi}.}]
  \item All major West Coast AM stations (Class~A and Class~B, PNW and
    California) individually documented with founding date, founding
    licensee, call sign history, frequency assignment history, network
    affiliations, and format evolution.
  \item AM band structure fully explained: frequency range, channel
    classification (clear/regional/local), station classes (A through D),
    power limits, and daytime vs.\ nighttime propagation physics.
  \item FCC licensing pathways for AM, FM, LPFM, FM translator, and Part
    ~15 operation described with step-by-step process, required forms,
    filing fees, and eligibility requirements.
  \item Spectrum scarcity context for Seattle, Portland, San Francisco,
    and Los Angeles markets quantitatively documented with reference to
    current FCC allotment tables.
  \item 47 CFR Part~73 regulatory obligations summarized for new licensees
    : EAS, indecency rules, political broadcasting obligations, EEO
    requirements, and public inspection file requirements.
  \item Ownership restriction framework documented: local market numerical
     caps, cross-ownership rules, and changes since the Telecommunications
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    Act of 1996.
  \item AM digital transition options (HD Radio hybrid, all-digital MA3,
    FM translator) described with current adoption statistics and
    regulatory status.
  \item EV radio interference problem documented with technical basis and
    industry response (NAB, automakers, congressional action through the AM
     Radio for Every Vehicle Act).
  \item Don Lee Network and NBC Pacific Coast network organizational
    structure and West Coast station affiliations mapped with dates.
  \item Historical format evolution documented per era: network radio
    (1920s--1940s), Top~40 (1950s--60s), news-talk pivot (1970s--80s),
    consolidation era (1990s--present).
  \item All numerical claims (wattage, frequencies, station counts,
    audience shares) attributed to specific FCC documents, FCC databases,
    NAB, or peer-reviewed sources.
  \item Practical broadcaster entry guide synthesizes licensing, spectrum,
     and regulatory information into an actionable road map for a
    prospective West Coast broadcaster in 2026.
\end{enumerate}

\subsection{Relationship to Other GSD Documents}

\begin{center}
\rowcolors{2}{rowgray}{white}
\begin{tabularx}{\textwidth}{L{3.5cm}L{4cm}X}
\toprule
\rowcolor{primary}
\tableheader{Document} & \tableheader{Relationship} & \tableheader{Notes}
    \\
\midrule
FCC Part 73 Regulatory Reference & Informational dependency & Primary
    source for Modules 4 and 5 \\
Fox Infrastructure Group SPC Formation & Business context & Broadcasting
    license applicant entity context \\
GSD Ecosystem Core & Platform & Mission executed via gsd-skill-creator
    framework \\
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\end{tabularx}
\end{center}

\subsection{The Through-Line}

There is a reason the word ``broadcast'' means both a way of transmitting
    radio signals and the
act of scattering seed widely across a field. AM radio was the original
    broadcast medium---a
single transmitter flinging its signal across thousands of square miles,
    trusting that whoever
needed it would find it. It required nothing of its listeners but a
    receiver and attention.
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The West Coast, always a place where the frontier impulse found new
    expression, took to radio
with particular intensity. The Pacific Northwest's geography---deep fjords
    , mountain ranges,
vast distances between settlements---made radio not a luxury but a
    necessity. The same topography
that makes the Cascades beautiful makes line-of-sight communication
    impossible. AM's ability to
follow the curve of the earth, to refract off the ionosphere at night and
    reach Nome or Nome
reach Seattle, was not just technically useful; it was democratically
    profound.

The builder who picks up this document in 2026 inherits that
    infrastructure. The frequencies
are still there, allocated and waiting. The regulations are complex but
    navigable. The question
is not whether the dial has room for new voices---it always has---but
    whether those voices know
how to find their way onto it.

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% STAGE 2 — RESEARCH REFERENCE
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\stageheader{STAGE 2 --- RESEARCH REFERENCE}{secondary}

\section{AM Radio: West Coast \& PNW --- Research Reference}

\subsection{How to Use This Document}

This reference compiles primary and secondary research across all six
    modules. Each section
provides substantive findings with source attribution. Agents should treat
     this as the ground
truth for the module they are producing; supplement with additional
    research only where
specifically noted as incomplete.

\textbf{Key source organizations:}
\begin{itemize}[leftmargin=*]
  \item Federal Communications Commission (FCC) --- fcc.gov (primary
    regulatory authority)
  \item eCFR / Code of Federal Regulations --- Title 47, Part 73
  \item California Historical Radio Society --- californiahistoricalradio.
    com
  \item HistoryLink.org --- Washington State history archive
  \item Bay Area Radio Museum \& Hall of Fame --- bayarearadio.org
  \item National Association of Broadcasters (NAB)
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  \item Prometheus Radio Project / Common Frequency (LPFM advocacy)
  \item Radio Survivor --- radiosurvivor.com
  \item World Radio History Archive --- worldradiohistory.com
\end{itemize}

% —— MODULE 1: AM BAND FOUNDATIONS
   ————————————————————————————————————————————

\subsection{Module 1 Reference: AM Band Foundations}

\subsubsection{The Physics of AM and Band Boundaries}

The AM broadcast band occupies 535 to 1,705~kHz (formerly 535--1,600~kHz,
   expanded in the 1990s
after years of international negotiation). AM channels are spaced 10~kHz
   apart, yielding
approximately 117 usable channels. The band was expanded upward to 1,700~
   kHz by the FCC in the
1990s, with frequencies above 1,600~kHz (the ``expanded band'') reserved
   for existing stations
that caused significant interference lower in the band.

AM propagation differs fundamentally between day and night. During
   daylight hours, the signal
travels primarily by ground wave, with a reliable range of up to 200 miles
    depending on power.
At night, the ionosphere's D layer dissipates, allowing the signal to
   reflect off the E and F
layers and skip hundreds to thousands of miles. This ``skywave''
   propagation is why nighttime AM
can be heard across continents---and why stations sharing a frequency must
    coordinate power and
antenna directionality to prevent interference.

\subsubsection{Regulatory History: FRC, FCC, and the Radio Acts}

Before 1927, the AM dial was chaos. Hundreds of stations operated at
   overlapping frequencies
with no mandatory coordination. The Federal Radio Commission (FRC) was
   established by the Radio
Act of 1927, which for the first time required stations to operate in the
   public interest,
convenience, and necessity. The FRC implemented General Order 40 in
   November 1928, introducing
the ``clear channel'' concept and reassigning stations to rationalized
   frequencies.

The Communications Act of 1934 replaced the FRC with the Federal
   Communications Commission
(FCC), which has regulated broadcast spectrum ever since. The North
   American Regional Broadcasting
Agreement (NARBA) of 1941 coordinated AM frequency assignments between the
    United States, Canada,
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and Mexico, shifting many West Coast stations to new dial positions (e.g.,
    KOMO shifted from
950 to 1000~kHz under NARBA).

\subsubsection{AM Station Classification System}

\begin{center}
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\begin{tabularx}{\textwidth}{L{2cm}L{3cm}XL{3.5cm}}
\toprule
\rowcolor{primary}
\tableheader{Class} & \tableheader{Old Class} & \tableheader{Power /
   Coverage} & \tableheader{Key Protection} \\
\midrule
Class A & Class I-A / I-B & Up to 50,000 W; nationwide nighttime & Highest
   ; clear-channel domination \\
Class B & Class II & Up to 50,000 W; regional & Protected from Class A
   interference \\
Class C & Class III & Up to 1,000 W; local & Local market only \\
Class D & Class IV & Daytime only; up to 1,000 W at night conditionally &
   Sunset-to-sunrise silent or power reduction \\
\bottomrule
\end{tabularx}
\end{center}

Class~A stations dominate the AM dial at night. The FCC currently
   estimates 4,400 licensed
commercial AM stations in the United States as of September 2024 (Federal
   Register, March 2025).

% ── MODULE 2: PNW HISTORICAL STATIONS
   ────────────────────────────────────────────────

\subsection{Module 2 Reference: Pacific Northwest Historical Stations}

\subsubsection{The First Wave: Experimental Stations, 1920--1925}

Radio came to the Pacific Northwest almost simultaneously with its
   commercial emergence nationwide.
The first convention of the Northwestern Radio Association was held in
   Portland on June 25, 1921,
attended by 125 members from amateur radio clubs across Washington, Oregon
   , Idaho, and Northern
California---at a time when only KDKA in Pittsburgh held a commercial
   license. Portland's Charles
L.~Austin (call sign 7XF) had been broadcasting phonograph music to ships
   in the Portland Harbor
from his home laboratory on Mount Tabor, and West Coast inventors were
   already frustrated that
eastern interests received more regulatory recognition.

KJR Seattle, begun by amateur radio operator Vincent~I.~Kraft, was the
   first radio station
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licensed in the Pacific Northwest. It went on the air as an experimental
    station in 1921 and was
one of the first commercial licenses in the region. In October 1926, Kraft
     co-founded the Pacific
Broadcasting Corporation with Frederick~C.~Clift of San Francisco, which
    created KYA in San
Francisco using the call letters and a new transmitter---a direct business
     connection between
Seattle and Bay Area broadcasting from the earliest days.

KGB in Tacoma went on the air on April 4, 1922, operated by the Tacoma
    Daily Ledger and the
William~A.~Mullins Electric Company. Notably, its radio editor was
    Winifred Dow, one of the few
women in the Pacific Northwest to hold an amateur radio license in that
    era.

\subsubsection{Key PNW AM Stations: Profiles}

\begin{center}
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\begin{tabularx}{\textwidth}{L{2.2cm}C{1.5cm}L{2.5cm}X}
\toprule
\rowcolor{primary}
\tableheader{Station} & \tableheader{Freq.} & \tableheader{Founded} & \
    tableheader{Key History} \\
\midrule
KJR Seattle & 950 & 1921 & First PNW license; NBC Blue; ESPN Radio;
    Classic Rock 1988 \\
KNWN/KOMO & 1000 & 1926 & 50 kW Class A; NBC; Fisher family; Mariners; now
     Lotus Communications \\
KIRO Seattle & 710 & 1927 & CBS affiliate; all-news; Bonneville; 50 kW
    Class B \\
KVI Seattle & 570 & 1926 & Sports; personality radio; news 1980; Mariners
    pre-KOMO \\
KGW Portland & 620 & 1922 & NBC affiliate; Orange Chain; first PNW TV 1952
     \\
KOIN Portland & 940 & 1928 & Don Lee/CBS affiliate; longtime Portland
    institution \\
KEX Portland & 1190 & 1927 & NBC Gold Network; Pacific Gold Chain station
    \\
KHQ Spokane & 590 & 1922 & Spokane dominant; NBC; one of earliest 50 kW
    outlets \\
KGA Spokane & 1510 & 1927 & NBC Gold; regional powerhouse \\
KFAE Pullman & --- & 1922 & Educational; Washington State College; first
    edu. station in WA \\
\bottomrule
\end{tabularx}
\end{center}

\subsubsection{The Adolph Linden Empire and Early Consolidation}
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Adolph Linden and Edmund Campbell, directors of Puget Sound Savings \&
    Loan, founded Northwest
Radio Service in 1926 and acquired KJR from Vincent Kraft. Linden then
    created the American
Broadcasting Company (ABC)---not the later national network---which by
    1929 included stations
in Portland, Spokane, San Francisco, and ``a half-dozen others from here
    to Chicago.''

This early consolidation foreshadowed the industry's later corporate
    concentration, and Linden's
story ended badly: he was convicted of embezzlement from Puget Sound
    Savings \& Loan in 1932
and served five years in the Walla Walla state penitentiary.

\subsubsection{NBC Comes to Seattle: The Network Era, 1926--1950}

In 1926, special AT\&T wires reached Seattle from New York, and KOMO
    became an affiliate of NBC,
the National Broadcasting Company. Local listeners could now hear the same
     programs---and the
same national commercials---as people across the country. NBC organized
    its West Coast operations
into the ``Pacific Coast Network'' with Red and Blue designations. In
    1931, NBC reorganized into
the Pacific Gold Network (KEX Portland; KJR Seattle) and the Pacific
    Orange Network (KFI Los
Angeles; KGO Oakland; KGW Portland; KOMO Seattle; KHQ Spokane).

On September 26, 1937, the Mutual Broadcasting System made its Northwest
    debut with a 90-minute
inaugural program. The Don Lee Network, which produced programming for
    Mutual's West Coast
affiliates, included: KOL Seattle, KALE Portland (KPOJ), KMO Tacoma, KORE
    Eugene, KSLM Salem,
KIT Yakima, KVOS Bellingham, KGY Olympia, KIEM Eureka, KPQ Wenatchee, KRNR
     Roseburg, and KXRO
Aberdeen---described at the time as the largest regional network in the
    United States.

\subsubsection{Format Evolution: Top 40, News-Talk, and the FM Defection}

Top~40 radio arrived in the Pacific Northwest in the 1950s, and KJR
    Seattle became one of the
most dominant personality stations in the region through the 1960s and
    1970s. KVI (570~AM)
carried the pre-Mariners Seattle Rainiers baseball and was a powerhouse
    personality station.
When music fans migrated to FM's superior sound in the 1970s, AM pivoted
    to news and talk.
KIRO became an all-news powerhouse; KOMO followed with news-talk by 1995.
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KJR moved to Classic Rock in 1988, then Sports/ESPN in 1992---a format it
    held for decades.
KOMO's call letters migrated to KNWN (Northwest News Radio) in February
    2022 after Lotus
Communications acquired the Fisher/Sinclair radio properties.

% —— MODULE 3: CALIFORNIA HISTORICAL STATIONS
    ————————————————————————————————————

\subsection{Module 3 Reference: California Historical Stations}

\subsubsection{Origins: The World's First Broadcaster?}

Charles Herrold of San Jose was broadcasting music and talk on a regular
    schedule between 1912
and 1917, to a growing audience and to receiving stations at the 1915
    Panama-Pacific International
Exhibition. His station, which became KCBS (740~AM), is one of the
    strongest candidates for the
title of the world's first continuously operating broadcast station---a
    claim that predates
KDKA Pittsburgh's November 1920 commercial debut. Following World War~I's
    radio silence, it
reemerged and eventually became a CBS-owned-and-operated news powerhouse.

In April 1920, the Western Radio Electric Company received a license for
    experimental station
6XD in Los Angeles (920~kHz), described by Pacific Radio News in 1921 as
    broadcasting on 325
meters two nights a week. This was the first broadcasting station in
    Southern California, active
approximately five months before KNX's predecessor, 6ADZ.

\subsubsection{Key California AM Stations: Profiles}

\begin{center}
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\begin{tabularx}{\textwidth}{L{2.5cm}C{1.5cm}L{2.5cm}X}
\toprule
\rowcolor{primary}
\tableheader{Station} & \tableheader{Freq.} & \tableheader{Founded} & \
    tableheader{Key History} \\
\midrule
KGO/KSFO SF & 810 & Jan 8, 1924 & GE/RCA/NBC Blue/ABC; Jack LaLanne; 50 kW
     Class A; call changed to KSFO in 2024 \\
KNBR SF & 680 & 1922 & NBC Red flagship SF; sports powerhouse; Giants/
    Warriors \\
KYA SF & 1260 & Dec 17, 1926 & Pacific Broadcasting Corp; CBS affiliate
    1928 \\
KCBS SF & 740 & 1909* & World's first continuous broadcaster (Herrold);
    CBS O\&O; news \\
KFI Los Angeles & 640 & Apr 16, 1922 & First 50 kW west of Chicago (1931);
     Class A; Earle C. Anthony/NBC \\
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KNX Los Angeles & 1070 & 1920 & Westinghouse/CBS O\&O; AP/UPI; Breaking
   News Murrow Award 2017 \\
KABC Los Angeles & 790 & 1925 & NBC Blue/ABC; news-talk; iHeartMedia \\
KMPC Los Angeles & 710 & 1922 & Music of Your Life; sports; later KSPN \\
\bottomrule
\end{tabularx}
\end{center}

\subsubsection{KGO/KSFO: The West Coast Clear-Channel Giant}

KGO signed on January 8, 1924, from General Electric's Oakland transformer
    plant as part of a
planned three-station GE network (with WGY Schenectady and KOA Denver). It
    was known as the
``Sunset Station.'' Due to GE's involvement with RCA, KGO was soon
   operated under NBC management.
The 1928 Federal Radio Commission reallocation designated clear-channel
   status on 810~kHz
originally at 790~kHz; NARBA in 1941 moved it to 810.

In 1943, the FCC forced NBC to sell one of its two networks; the Blue
   Network became ABC, and
KGO became ABC's West Coast flagship. In the postwar period, KGO produced
   live Western Swing
music and hosted the first radio exercise show by Jack LaLanne.

By the 1960s KGO had reinvented itself as one of America's first news-talk
    stations under Program
Director Jim Dunbar, who broadcast live from local restaurants and
   nightclubs. It held Arbitron's
\#1 Bay Area ranking for over 27 consecutive years. In 2024, the ``KGO''
   call letters were retired
and the station became KSFO---a notable end to a century-long call sign
   history.

\subsubsection{KFI: The First 50,000-Watt Station West of Chicago}

KFI (640~AM) was first licensed March 31, 1922, by Earle~C.~Anthony, owner
    of a Packard
automobile dealership who had trained as an electrical engineer. It made
   its debut broadcast
on April 16, 1922, with vaudeville performers Eugene and Willie Howard.

In July 1931, KFI increased its transmitter power to 50,000 watts---the
   first station west of
Chicago to broadcast at that power. A special four-hour program marked the
    occasion, with
congratulatory speeches joined by NBC entertainers on a coast-to-coast
   hookup. KFI was part
of the NBC Pacific Orange Network alongside KGO, KGW, KOMO, and KHQ.
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Today KFI (owned by iHeartMedia) remains one of the highest-rated talk
    stations on the West
Coast, with Bill Handel's morning show running since 1993. Its signal
    covers Southern California
by day and reaches much of western North America at night.

\subsubsection{The Don Lee Network and California Infrastructure}

Don Lee, a Cadillac dealer turned broadcaster, built the West Coast's most
     extensive broadcast
infrastructure in the 1930s. The Don Lee Network produced programming for
    the Mutual Broadcasting
System and served as its West Coast backbone. Don Lee stations included
    KFRC San Francisco,
KHJ Los Angeles, and affiliates throughout California and into the Pacific
     Northwest. Don Lee
became a stockholder in Mutual in 1940.

% —— MODULE 4: SPECTRUM AND LICENSING
    ————————————————————————————————————————

\subsection{Module 4 Reference: Spectrum Availability and Licensing
    Pathways}

\subsubsection{The Spectrum Scarcity Reality}

The FCC states explicitly: ``In many areas of the country, no frequencies
    may be available on
which a new station could begin operating without causing interference to
    existing stations.''
This is especially acute in the top 50 markets---Seattle, Portland, San
    Francisco, Los Angeles,
and San Diego---where the FM dial is heavily occupied and AM frequencies
    are constrained by
Class~A clear-channel protections and NARBA international obligations.

The FM band (88.1--107.9~MHz) cannot be expanded: aeronautical navigation
    operations occupy
108--136~MHz above, and Channel~6 TV operations (82.0--88.0~MHz) constrain
     the lower edge.
The AM band was last expanded in the 1990s from 1600 to 1705~kHz, and
    further expansion is
described by the FCC as unlikely.

\subsubsection{Licensing Pathway Comparison}

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\toprule
\rowcolor{primary}
\tableheader{Type} & \tableheader{Key Requirements} & \tableheader{Power /
     Range} & \tableheader{Availability} \\
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\midrule
Full-Power AM & Citizenship, character, technical engineering study,
   construction permit, FCC Form~301 & 250 W minimum; Class A up to 50 kW
   & Very limited in major markets \\
Full-Power FM & Engineering study, open allotment, FCC Form~301,
   application fee & 100 W minimum (Class A) to 100 kW (Class C) & Very
   limited in major markets \\
LPFM & Nonprofit/educational entity; community-based (10-mile rule); FCC
   Form 318 during filing window & Up to 100 W; 3.5 mile radius & Filing
   windows \textasciitilde{}every 10 years; last window Nov--Dec 2023 \\
FM Translator & Rebroadcasts another station; FCC Form 349 & Typically
   1--250 W & Limited; AM Revitalization opened access \\
Part 15 Unlicensed & No license required; must accept all interference &
   200-foot effective radius & Always available; minimal coverage \\
\bottomrule
\end{tabularx}
\end{center}

\subsubsection{LPFM: The Community Broadcaster's Pathway}

The LPFM service was created by the FCC in January 2000. LPFM stations are
    authorized for
noncommercial educational broadcasting only (no commercial operation) and
   operate with up to
100 watts ERP at 30 meters antenna height, with an approximate service
   radius of 3.5 miles
(5.6~km). LPFM stations are not protected from interference from full-
   power stations.

Eligibility is restricted to nonprofits and governmental/educational
   entities that are
community-based (physically headquartered or 75\% of board members within
   10 miles of the
proposed transmitter; 20 miles outside the top 50 markets). Individuals
   and commercial entities
are ineligible. Current broadcasters, cable operators, and newspaper
   publishers are also ineligible.

Applications are only accepted during FCC-announced filing windows. As of
   early 2026, the FCC
is NOT accepting new LPFM applications. The most recent window opened
   November 1, 2023, and
closed (revised) December 13, 2023---the third window ever, following
   2000--2001 and 2013.
As of March 2023, there were 1,999 licensed LPFM stations in the United
   States. The next
window is expected approximately 10 years hence.

There is no filing fee for LPFM construction permit applications. Station
   construction and
operating costs vary; a basic LPFM setup can range from a few thousand to
   tens of thousands
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of dollars depending on transmitter, antenna, tower, and studio
    requirements.

\subsubsection{Full-Power AM and FM: The Long Road}

For a commercial full-power AM station, applicants must: (1) identify an
    available frequency
using FCC AM Query tools; (2) commission a broadcast engineering study
    demonstrating no
interference to existing stations; (3) file FCC Form~301 (construction
    permit) electronically
via the Licensing and Management System (LMS) during an application window
    ; (4) pay the
applicable filing fee; and (5) await FCC processing, which may involve
    comparative hearings
if competing applications are filed.

FCC licenses are issued for eight-year terms. The minimum AM power is 250
    watts. Applications
must be filed electronically---paper forms are not accepted, though
    printed forms may be used
to prepare data. All applicants must be registered in the FCC's COmmission
     REgistration System
(CORES) with a valid FCC Registration Number (FRN).

\subsubsection{AM Revitalization: FM Translators for AM Stations}

As part of the ongoing AM Revitalization proceeding (Docket 13-249), the
    FCC has allowed AM
stations to obtain FM translator licenses to rebroadcast their programming
     on the FM band.
More than 2,100 FM translators were on the air rebroadcasting AM signals
    by 2020, with
approximately 700 more pending. This effectively allows daytime-only AM
    stations to maintain
a 24-hour presence on FM.

% ── MODULE 5: FCC RULES AND REGULATIONS
    ────────────────────────────────────────

\subsection{Module 5 Reference: FCC Rules and Regulations}

\subsubsection{The Regulatory Framework: 47 CFR Part 73}

All broadcast station rules are contained in 47 CFR Part~73, divided into
    subparts:
\begin{itemize}[leftmargin=*]
  \item \textbf{Subpart A} --- AM Broadcast Stations (\S\S~73.1--73.190)
  \item \textbf{Subpart B} --- FM Broadcast Stations (\S\S~73.201--73.333)
  \item \textbf{Subpart C} --- Digital Audio Broadcasting (\S\S
    ~73.401--73.406)
  \item \textbf{Subpart D} --- Noncommercial Educational FM (\S\S
    ~73.501--73.599)
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  \item \textbf{Subpart G} --- Low Power FM Broadcast Stations (\S\S
    ~73.801 et seq.)
  \item \textbf{Subpart H} --- Rules Applicable to All Broadcast Stations
    (\S\S~73.1001 et seq.)
\end{itemize}

Subpart~H is the licensee's primary compliance reference, covering
    political broadcasting,
EEO, EAS, indecency, and public file requirements.

\subsubsection{Public Interest Obligations}

The spectrum belongs to the public. Under the Radio Act of 1927 and the
    Communications Act
of 1934, licensees hold no property rights to the spectrum---only the
    privilege of using it
so long as they serve the public interest, convenience, and necessity. The
     FCC may revoke a
license before its term expires with notice and opportunity to be heard.

Key obligations for all broadcast licensees:
\begin{itemize}[leftmargin=*]
  \item \textbf{Equal Employment Opportunity (EEO):} Stations with five or
     more full-time employees
        must maintain outreach programs for employment and file FCC Form
    ~396 (EEO Report) at
        license renewal.
  \item \textbf{Political Broadcasting:} During candidate elections,
    stations must provide equal
        time (``equal opportunity'') to legally qualified candidates for
    the same office under
        47 U.S.C.~\S~315; must offer lowest unit rate during campaign
    windows.
  \item \textbf{Emergency Alert System (EAS):} All AM and FM stations must
     participate in the EAS,
        which requires FCC-compliant EAS decoders/encoders. KFI and KNX
    serve as primary EAS
        entry points for the Southern California Emergency Alert System.
  \item \textbf{Indecency and Obscenity:} 18 U.S.C.~\S~1464 prohibits
    obscene, indecent, and
        profane broadcasts. Indecent content (material that depicts or
    describes sexual or
        excretory organs in a patently offensive manner) is prohibited
    between 6:00~a.m.\ and
        10:00~p.m. local time. Obscene content is prohibited at all times.
  \item \textbf{Public Inspection File:} Stations must maintain a publicly
     accessible online
        file (OPIF) through the FCC containing the license, applications,
    EEO records, political
        file, and other documents. LPFM stations are exempt from the local
     public file requirement
        but must maintain a political file.
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  \item \textbf{On-Air Announcements:} During license renewal, stations
    must make on-air
        announcements about their renewal application within five business
     days of FCC public
        notice that the application has been accepted for filing.
\end{itemize}

\subsubsection{License Terms and Renewal}

FCC broadcast licenses are issued for eight-year terms. All radio station
    licenses are scheduled
to expire between 2027 and 2030, on a state-by-state schedule. Renewal
    applications (FCC
Form~303-S) must be filed four months before the expiration date. The FCC
    may not automatically
renew a license; the public may file Petitions to Deny or informal
    objections.

\subsubsection{Ownership Rules}

The Telecommunications Act of 1996 eliminated national radio station
    ownership caps. Local
ownership is governed by numerical market tiers based on total commercial
    stations in a market:
\begin{itemize}[leftmargin=*]
  \item In markets with 45+ stations: one entity may own up to 8 stations
    (no more than 5 in one service, AM or FM)
  \item In markets with 30--44 stations: up to 7 (4 per service)
  \item In markets with 15--29 stations: up to 6 (4 per service)
  \item In markets with 14 or fewer: up to 5 (3 per service)
\end{itemize}

The FCC's cross-ownership rules and the dual-network rule (\S~73.658)
    further constrain media
consolidation. The dual-network rule prohibits common ownership of two of
    the ``top four''
networks (ABC, CBS, Fox, NBC).

\subsubsection{Technical Standards}

AM stations must use FCC-approved transmitting equipment and must operate
    during hours
appropriate to their frequency class. Class~D (daytime) stations must
    cease or reduce power
at local sunset. Directional antenna arrays require FCC approval of the
    antenna system.
The minimum AM power for a construction permit is 250 watts.

The March 2025 FCC rulemaking (MB Docket 24-626) proposed updates to Part
    ~73 rules to
reflect current application processing requirements, update form names,
    remove CDBS references
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in favor of the LMS system, and eliminate certain AM power increase
    restrictions.

% —— MODULE 6: DIGITAL FUTURE
    ————————————————————————————————————————————————————

\subsection{Module 6 Reference: The Digital Future of AM Radio}

\subsubsection{The AM Revitalization Proceeding (Docket 13-249)}

In October 2013, the FCC issued a Notice of Proposed Rulemaking titled ``
    Revitalization of the
AM Radio Service.'' The proceeding has produced multiple Report and Orders
    , including:
\begin{itemize}[leftmargin=*]
  \item \textbf{First R\&O (2015):} Changes to daytime/nighttime community
     coverage standards;
        elimination of the AM ``ratchet'' rule; easier MDCL (Modulation
    Dependent Carrier Level)
        requirements; relaxation of minimum antenna efficiency standards.
  \item \textbf{Second R\&O:} FM translator access for AM stations.
  \item \textbf{Third R\&O (2017):} Relaxation of directional antenna
    array requirements.
  \item \textbf{Auction 99 (2018):} FM translator auction specifically for
     AM stations.
  \item \textbf{All-Digital R\&O (2020):} Authorized voluntary all-digital
     AM operation.
\end{itemize}

\subsubsection{HD Radio and All-Digital AM}

HD Radio (In-Band On-Channel, or IBOC digital) operates as a hybrid system
    , adding digital
sidebands to the existing analog AM carrier. The FM digital power increase
     order approved
by the FCC in 2010 and the asymmetric digital sideband order of September
    2024 have expanded
HD Radio capabilities on FM. On AM, the HD Radio hybrid has faced adoption
     challenges.

In December 2020, the FCC approved voluntary all-digital AM operation
    using the MA3 mode
(the HD Radio standard for all-digital). This allows AM stations to cease
    analog transmission
and broadcast entirely in digital, improving audio quality and signal
    robustness. Hubbard
Broadcasting's WWFD in Frederick, Maryland, conducted the landmark
    experiments; testing showed
reliable daytime coverage to the 0.5~mV predicted contour and nighttime
    coverage to the NIF
contour. As of 2020, approximately 95 million vehicles on U.S. roads were
    equipped with HD
Radio receivers.
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\subsubsection{The EV Interference Crisis}

Electric vehicles generate electromagnetic interference that degrades AM
    reception. Automakers
including Ford, BMW, and Volkswagen began removing AM radio from some EV
    models in 2023.
Ford initially announced removal from 2024 models; after consumer,
    broadcaster, and congressional
backlash, Ford CEO Jim Farley reversed the decision. The National
    Association of Broadcasters
urged all automakers to maintain AM capability. Congressional activity
    produced the AM Radio
for Every Vehicle Act, advancing through committee in 2023--2024.

GM and Stellantis announced no plans to remove AM from their vehicles,
    while keeping product
planning details confidential. The underlying technical problem---EV motor
     and charging
system emissions in the AM frequency range---remains unresolved.

\subsubsection{The Competitive Landscape}

As of March 2024, there were 17,370 licensed terrestrial stations in the
    United States (FCC,
Communications Marketplace Report 2024). AM/FM radio's combined share of
    total audio listening
fell to 36\% in 2023 (Edison Research), while online audio platforms
    gained ground. Terrestrial
radio revenue reached \$15.2 billion in 2023, down from \$17.8 billion in
    2019.

In-car listening remains AM/FM radio's primary stronghold. Monthly podcast
     listenership rose
to 47\% of the U.S. population in 2024, with podcast ad revenue reaching \
    $1.9 billion in 2023.
Spotify leads audio streaming with 35\% of U.S. listeners aged 12+.
    SiriusXM had 33.9 million
subscribers in 2023 (\$6.8 billion revenue).

Despite these pressures, AM broadcasting serves critical roles that
    streaming cannot replicate:
emergency alert infrastructure, hyperlocal community service in remote
    areas, and minority-language
programming. In California alone, 34\% of AM stations serve specific
    minority communities (as
documented in the AM Revitalization proceeding comments).

\subsubsection{The Path Forward: Practical Options for AM Stations}

\begin{enumerate}[leftmargin=*, label=\textbf{\arabic*.}]
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  \item \textbf{All-digital conversion:} File FCC Form~335-AM Digital
   Notification; commence
        all-digital operation 30 days after FCC public notice. Requires
   transmitter capable of MA3.
  \item \textbf{FM translator acquisition:} File for an FM translator to
   simulcast AM content on FM,
        providing 24/7 coverage even for daytimer stations.
  \item \textbf{HD Radio hybrid:} Deploy IBOC digital sidebands on
   existing AM analog authorization.
  \item \textbf{Internet simulcast:} Stream AM programming online at no
   additional FCC licensing cost.
  \item \textbf{Emergency infrastructure positioning:} Emphasize EAS role
   and community resilience
        value to justify ongoing public interest obligations.
\end{enumerate}

\subsection{Safety and Sensitivity Considerations}

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\tableheader{Area} & \tableheader{Consideration} & \tableheader{Boundary
   Type} \\
\midrule
Source quality & All claims must cite FCC documents, peer-reviewed
   research, or professional broadcast organizations. No entertainment
   media, blogs, or unsourced claims. & ABSOLUTE \\
Numerical attribution & Every wattage, frequency, station count, or
   audience share attributed to specific source. & ABSOLUTE \\
No policy advocacy & Document presents regulatory information and findings
    without advocating for specific FCC policy outcomes. & ABSOLUTE \\
Legal advice boundary & Regulatory summaries are informational only;
   applicants must consult a broadcast attorney for specific licensing
   advice. & ANNOTATE \\
Minority programming & Statistics on minority-language and minority-
   community AM stations presented factually and without characterization.
    & GATE \\
\bottomrule
\end{tabularx}
\end{center}

\subsection{Source Bibliography}

\subsubsection{Government and Agency Sources}
\begin{itemize}[leftmargin=*]
  \item Federal Communications Commission. \textit{How to Apply for a
   Radio or Television Broadcast Station.} fcc.gov/media/radio/how-to-
   apply
  \item Federal Communications Commission. \textit{Low Power FM (LPFM)
   Broadcast Radio Stations.} fcc.gov/media/radio/lpfm
```

```
  \item Federal Communications Commission. \textit{AM Revitalization.} fcc
    .gov/media/radio/am-revitalization
  \item Federal Communications Commission. \textit{Digital Radio.} fcc.gov
    /media/radio/digital-radio
  \item Federal Communications Commission. \textit{AM Station Classes, and
     Clear, Regional, and Local Channels.} fcc.gov/media/radio/am-clear-
    regional-local-channels
  \item Federal Communications Commission. \textit{All-Digital AM
    Broadcasting, Revitalization of the AM Radio Service.} FCC 20-154 (
    Report and Order, Dec.\ 2020). Federal Register, Dec.\ 3, 2020.
  \item Federal Communications Commission. \textit{Broadcast Station Rule
    Updates.} MB Docket 24-626. Federal Register, Mar.\ 24, 2025.
  \item eCFR. \textit{47 CFR Part 73 --- Radio Broadcast Services.} ecfr.
    gov/current/title-47/chapter-I/subchapter-C/part-73
  \item FCC. \textit{Communications Marketplace Report 2024.} (Cited in
    Radio Ink, Jan.\ 7, 2025)
\end{itemize}

\subsubsection{Professional Organizations and Research}
\begin{itemize}[leftmargin=*]
  \item California Historical Radio Society. \textit{100 Years of Radio.}
    californiahistoricalradio.com (2011, updated 2024)
  \item HistoryLink.org. Banel, Feliks. \textit{Radio in Washington.}
    historylink.org/File/22494
  \item Bay Area Radio Museum \& Hall of Fame. \textit{KYA/KOIT Historical
     Timeline.} bayarearadio.org
  \item Radio Survivor. \textit{Radio Recollections: The San Francisco Bay
     Area in the 1950s.} radiosurvivor.com (June 2016)
  \item Common Frequency. \textit{Overview on Applying for Low Power FM in
     2023.} commonfrequency.org
  \item Xperi. \textit{Is AM Radio Dying? Not Any Time Soon.} xperi.com/
    blog (2023)
  \item Radio Ink. \textit{FCC Captures Radio's Struggle Against Digital
    in Biennial Report.} radioink.com (Jan.\ 7, 2025)
  \item Radio World. \textit{FCC Approves All-Digital Option for AM.}
    radioworld.com (Oct.\ 27, 2020)
\end{itemize}

\subsubsection{Station Histories and Archives}
\begin{itemize}[leftmargin=*]
  \item Wikipedia contributors. \textit{KFI.} en.wikipedia.org/wiki/KFI (
    updated Feb.\ 2026)
  \item Wikipedia contributors. \textit{KSFO.} en.wikipedia.org/wiki/KSFO
    (updated 2026)
  \item Wikipedia contributors. \textit{KNWN (AM).} en.wikipedia.org/wiki/
    KNWN\_(AM) (updated Jan.\ 2026)
  \item QZVX.com. \textit{AM Radio Stations: PNW History.} qzvx.com/am-
    radio-stations/
  \item Swingin' West. \textit{Oldest Radio Stations / American / Radio
    History.} swinginwest.com
  \item Tripod / Jeff560. \textit{A Chronology of AM Radio Broadcasting
    1900--1960.} jeff560.tripod.com/chrono1.html
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  \item Oregon Radio / KGW / NBC History. ke7ggv.tripod.com/nwradiohistory
    .html
\end{itemize}

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\section{Milestone Specification}

\subsection{Mission Objective}

Produce a comprehensive, publication-quality research document on the
    history, present state,
and future of AM radio broadcasting on the West Coast of the United States
    , with deep focus on
the Pacific Northwest and California. The document will serve both as an
    authoritative historical
reference and as a practical licensing and regulatory guide for
    prospective broadcasters in 2026.
All claims will be sourced to FCC documents, professional broadcast
    organizations, or peer-reviewed
research.

\subsection{Architecture Overview}

\begin{asciibox}
WAVE EXECUTION ARCHITECTURE
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  WAVE 0: FOUNDATION
  ┌──────────────────────────────────────────────────────────────┐
  │  W0-A: Shared Schema (Haiku)   W0-B: Source Index (Sonnet)│
  └──────────────────────────────────────────────────────────────┘
                                 │
  WAVE 1: PARALLEL RESEARCH TRACKS
  ┌──────────────────────────────────────────────────────────────┐
  │  TRACK ALPHA                  │  TRACK BETA                  │
  │  W1-A: AM Band Foundations    │  W1-D: Spectrum & │Licensing
  │  W1-B: PNW Station History    │  W1-E: FCC Regulations       │
  │  W1-C: California Stations    │                              │
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                                 │
  WAVE 2: SYNTHESIS
  ┌──────────────────────────────────────────────────────────────┐
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   │  W2-A: Cross-Module Integration (Opus)                    │
   │  W2-B: Digital Future Analysis (Opus)                     │
   └──────────────────────────────────────────────────────────┘
                                 │
  WAVE 3: PUBLICATION + VERIFICATION
   ┌──────────────────────────────────────────────────────────┐
   │  W3-A: Document Assembly (Sonnet)                         │
   │  W3-B: Source Verification (Sonnet)                       │
   │  W3-C: CAPCOM Review Gate (Human)                         │
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                      FINAL DELIVERABLE PDF
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1.6 System Layers

1. **Foundation Layer** — Shared data schemas, source index, station database structure
2. **Historical Research Layer** — Station-by-station profiles, network chronologies, format evolution
3. **Regulatory Research Layer** — FCC rule summaries, licensing pathways, spectrum analysis
4. **Synthesis Layer** — Cross-module integration, digital future analysis, practical broadcaster guide
5. **Publication Layer** — Document assembly, formatting, source verification, final PDF

1.7 Deliverables

#	Deliverable	Acceptance Criteria	Primary Spec
1	Station History Compendium	All Class A/B PNW and CA stations profiled; dates, frequencies, affiliations verified	W1-B, W1-C
2	AM Band Technical Reference	Band structure, propagation physics, station classes, NARBA explained	W1-A
3	Licensing Pathway Guide	All five pathways documented; step-by-step; current as of 2026	W1-D
4	FCC Regulatory Summary	47 CFR Part 73 obligations for new licensees; ownership rules; EAS	W1-E
5	Digital Transition Analysis	HD Radio, all-digital AM, FM translators, EV interference, market data	W2-B
6	Practical Broadcaster Road Map	Synthesized entry guide for prospective 2026 West Coast broadcaster	W2-A
7	Verified Source Bibliography	All sources traceable; no entertainment media	W3-B

1.8 Component Breakdown

Component	Dependencie	Model	Token Budget (K)
W0-A: Schema	None	Haiku	4
W0-B: Source Index	None	Sonnet	8
W1-A: AM Foundations	W0	Sonnet	18
W1-B: PNW Stations	W0, W1-A	Sonnet	28
W1-C: CA Stations	W0, W1-A	Sonnet	28
W1-D: Licensing	W0, W1-A	Sonnet	22
W1-E: FCC Rules	W0, W1-A	Sonnet	22
W2-A: Cross-Module Integration	W1-B, W1-C, W1-D, W1-E	Opus	35
W2-B: Digital Future	W1-A, W2-A	Opus	25
W3-A: Document Assembly	W2-A, W2-B	Sonnet	40
W3-B: Source Verification	W3-A	Sonnet	16
Total			**246**

1.9 Model Assignment Rationale

Assignment	Model	Rationale
Schema, scaffolding	Haiku	Mechanical, low-judgment template generation
Survey, compilation, verification	Sonnet	Standard research and document production
Cross-module synthesis, digital future analysis, broadcaster road map	Opus	Requires judgment, synthesis across 5+ modules, trend analysis and extrapolation

Model allocation: Opus ≈24% (60K tokens), Sonnet ≈72% (178K tokens), Haiku ≈4% (8K tokens).

2 Wave Execution Plan

2.1 Wave Summary

Wave	Name	Components	Mode	Gate
0	Foundation	W0-A, W0-B	Sequential	Auto
1	Parallel Research	W1-A through W1-E	Parallel (2 tracks)	CAPCOM
2	Synthesis	W2-A, W2-B	Sequential	CAPCOM
3	Publication	W3-A, W3-B, W3-C	Sequential	Human

2.2 Wave 0: Foundation

Task	Agent	Deliverable	Model
W0-A	PLAN + Haiku	Station database schema (JSON): call sign, frequency, city, class, founded, network affiliations, format history, current status	Haiku
W0-B	SCOUT + Sonnet	Annotated source index: URL, document type, reliability tier, module assignments for each of the 30+ sources gathered in cold-start research	Sonnet

2.3 Wave 1: Parallel Research Tracks

Track Alpha — History

Task	Agent	Deliverable	Model
W1-A	CRAFT-TECH	AM band foundations document: physics, FRC/FCC history, NARBA, station classification table, band expansion history	Sonnet
W1-B	CRAFT-HIST + EXEC-1	PNW station compendium: full profiles for all Class A/B stations in WA/OR/ID; network chronology; format evolution by decade	Sonnet
W1-C	CRAFT-HIST + EXEC-2	California station compendium: full profiles for Bay Area and LA Class A/B stations; Don Lee Network; NBC Pacific structures	Sonnet

Track Beta — Regulatory & Licensing

Task	Agent	Deliverable	Model
W1-D	CRAFT-REG	Licensing pathway guide: all five pathways with step-by-step, forms, fees, LPFM window schedule, spectrum scarcity context for 4 major West Coast markets	Sonnet
W1-E	CRAFT-REG	FCC regulatory summary: Part 73 subpart-by-subpart; public interest obligations; ownership rules; EAS; indecency; license terms; 2025 rule updates	Sonnet

2.4 Wave 2: Synthesis

Task	Agent	Deliverable	Model
W2-A	INTEG + Opus	Cross-module integration: consistent station data; practical broadcaster road map synthesizing history + licensing + regulations; editorial through-line	Opus
W2-B	ANALYST + Opus	Digital future analysis: HD Radio adoption trajectory; all-digital AM viability; EV interference technical analysis; competitive streaming landscape; 5-year AM outlook	Opus

2.5 Wave 3: Publication and Verification

Task	Agent	Deliverable	Model
W3-A	EXEC-1 + Sonnet	Full document assembly: LaTeX or structured markdown of all modules; tables, diagrams, bibliography formatted to publication standard	Sonnet
W3-B	VERIFY + Sonnet	Source verification report: every factual claim traced to source; SC-SRC and SC-NUM safety tests run; flagged items returned to producing agent	Sonnet
W3-C	CAPCOM	Human review gate: final HITL approval before PDF generation	Human

2.6 Cache Optimization Strategy

- Cache the annotated source index (W0-B) at session start for all Wave 1 agents—avoids 30+ URL lookups per agent
- Cache the AM band foundations document (W1-A) as shared context for all history and regulatory agents
- Maintain the station database schema (W0-A) in shared context throughout Wave 1 to ensure consistent data structure
- Batch all FCC LMS queries within a single SCOUT agent session in Wave 0 rather than distributing across agents
- Use prompt caching for the 47 CFR Part 73 regulatory text—accessed by both W1-D and W1-E

2.7 Token Budget

Wave	Opus (K)	Sonnet (K)	Haiku (K)	Wave Total (K)
Wave 0	0	8	4	12
Wave 1 (Alpha)	0	74	0	74
Wave 1 (Beta)	0	44	0	44
Wave 2	60	0	0	60
Wave 3	0	56	0	56
Total	**60**	**182**	**4**	**246**

2.8 Dependency Graph

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DEPENDENCY GRAPH═══════════════════════════════════════════════════════════

  [W0-A Schema] ──────────────────────────────────────────┐
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[W0-B Source Index] ──┬──────────────────────────────┤
                      │                              │
          ┌───────────┼──────────┬──────────┬        │
          ▼           ▼          ▼          ▼        ▼
      [W1-A]      [W1-B]     [W1-C]     [W1-D]   [W1-E]
      Foundations PNW Hist   CA Hist   Licensing  FCC Rules
          │           │          │          │        │
          └───────────┴──────────┴──────────┴────────┘
                                 │
                      ┌──────────┴──────────┐
                      ▼                     ▼
                  [W2-A]                 [W2-B]
                Cross-Module           Digital Future
                Integration              Analysis
                      │                     │
                      └──────────┬──────────┘
                                 │
                      ┌──────────┴──────────┐
                      ▼                     ▼
                  [W3-A]                 [W3-B]
                Document                Source
                Assembly              Verification
                      │                     │
                      └──────────┬──────────┘
                                 │
                             [W3-C]
                           CAPCOM GATE
                                 │
                        FINAL DELIVERABLE
```

3 Test Plan

3.1 Test Categories

Category	Count	Priority	Action	Target %
Safety-critical	7	Mandatory	BLOCK	16%
Core functionality	22	Required	BLOCK	49%
Integration	10	Required	BLOCK	22%
Edge cases	6	Best-effort	LOG	13%
Total	**45**			**100%**

3.2 Safety-Critical Tests

Test ID	Verifies	Expected
SC-SRC	Source quality	All citations traceable to FCC documents, peer-reviewed sources, or professional broadcast organizations. Zero entertainment media or unsourced blogs.
SC-NUM	Numerical attribution	Every wattage, frequency, station count, audience share, and date attributed to a specific named source.
SC-ADV	No policy advocacy	Document presents regulatory information without advocating for specific FCC policy changes or positions.
SC-LEG	Legal advice boundary	All regulatory summaries include disclaimer that content is informational and applicants must consult a broadcast attorney.
SC-MIN	Minority community data	Statistics on minority-language/community AM stations presented factually; no characterization or editorial framing.
SC-CUR	Temporal currency	Document correctly reflects 2026 regulatory status; no outdated CDBS references; LMS system referenced for filings.
SC-ELI	LPFM eligibility accuracy	LPFM eligibility restrictions (nonprofits only; no individuals; no commercial entities) stated accurately with CFR citation.

3.3 Core Functionality Tests (Selected)

Test ID	Verifies	Expected
CF-01	PNW station coverage completeness	All Class A and major Class B AM stations in WA/OR profiled with founding date, frequency, and ownership history
CF-02	California station completeness	All Class A AM stations in CA (KFI, KNX, KGO/KSFO, KNBR, KCBS, KABC) individually profiled
CF-03	Network chronology accuracy	NBC Pacific networks (Orange, Gold), Don Lee/Mutual, CBS, and ABC West Coast structures mapped with station affiliations and dates
CF-04	AM band structure documentation	535–1705 kHz range, 10 kHz channel spacing, Class A–D power limits, and daytime/nighttime propagation differences explained
CF-05	LPFM pathway accuracy	Filing window schedule, Form 318, eligibility requirements, 100 W power limit, 3.5-mile radius, and community-based requirement all documented
CF-06	Full-power AM pathway	Form 301, engineering study requirement, 250 W minimum, construction permit process, and FRN registration requirement documented
CF-07	FM translator pathway	AM Revitalization translator access, 2,100+ on-air translators statistic sourced to FCC, Form 349 referenced
CF-08	Part 15 unlicensed operation	200-foot effective radius, FCC Part 15 rules, absence of licensing requirement, and interference acceptance noted
CF-09	License term and renewal	Eight-year term, Form 303-S, four-month pre-filing deadline, 2027–2030 expiration schedule documented
CF-10	Ownership rules	1996 Telecom Act changes, local numerical market tiers, cross-ownership rules referenced
CF-11	EAS obligations	EAS decoder/encoder requirement, KFI/KNX roles documented, CFR citation included
CF-12	Indecency rules	18 U.S.C. § 1464, 6 a.m.–10 p.m. safe harbor, obscenity prohibition, and 47 CFR § 73.3999 all referenced
CF-13	All-digital AM authorization	FCC 20-154, MA3 mode, Form 335-AM Digital Notification, 30-day public notice window documented
CF-14	HD Radio adoption data	95 million vehicles with HD Radio sourced to Xperi/DTS 2023 data
CF-15	EV interference documented	Automaker removal decisions (Ford, BMW, VW), Ford reversal, congressional response, NAB position documented
CF-16	Audience share data	36% AM/FM share of audio listening in 2023 sourced to Edison Research via FCC 2024 report
CF-17	AM revitalization history	Docket 13-249 milestones (2013 NPRM through 2020 digital R&O) documented with dates
CF-18	NARBA frequency shifts	KOMO 950→1000, KGO 790→810 and other NARBA-driven reassignments documented
CF-19	KJR founding history	1921 founding by Vincent Kraft, first PNW license, KYA San Francisco connection documented
CF-20	KFI 50 kW milestone	July 1931 power increase, first west of Chicago, NBC Orange network affiliation documented
CF-21	KGO/KSFO	January 8, 1924 sign-on; GE/NBC/ABC lineage; Jack

3.4 Verification Matrix

Success Criterion	Test ID(s)	Status
SC-1. All Class A/B PNW and CA stations profiled	CF-01, CF-02, CF-03	Pending
SC-2. AM band structure fully explained	CF-04, CF-22	Pending
SC-3. FCC licensing pathways documented	CF-05, CF-06, CF-07, CF-08	Pending
SC-4. Spectrum scarcity context documented	CF-22, IN-01	Pending
SC-5. 47 CFR Part 73 obligations summarized	CF-09, CF-11, CF-12	Pending
SC-6. Ownership restrictions documented	CF-10	Pending
SC-7. AM digital transition options described	CF-13, CF-14	Pending
SC-8. EV interference documented	CF-15	Pending
SC-9. Don Lee / NBC Pacific networks mapped	CF-03, CF-21	Pending
SC-10. Historical format evolution documented	CF-17, CF-18, CF-19, CF-20	Pending
SC-11. All numerical claims attributed	SC-NUM, CF-16	Pending
SC-12. Practical broadcaster road map synthesized	IN-02, W2-A	Pending

4 Mission Crew Manifest

Role	Agent	Responsibility
FLIGHT	Orchestrator	Overall mission direction; go/no-go at each wave boundary
PLAN	Planner	Wave decomposition; parallel track optimization; dependency management
SCOUT	Research Agent	Source gathering; FCC database queries; web research; evidence compilation
EXEC-1	Executor Alpha	Document production for Track Alpha (PNW history, California history)
EXEC-2	Executor Beta	Document production for Track Beta (licensing, FCC rules)
CRAFT-HIST	History Specialist	Station-by-station chronicle; network affiliate research; archival sourcing
CRAFT-TECH	Technical Specialist	AM band physics; antenna systems; propagation; HD Radio technical standards
CRAFT-REG	Regulatory Specialist	47 CFR Part 73 analysis; FCC LMS procedures; ownership rules; EAS
VERIFY	Verification Agent	Source validation; SC-SRC and SC-NUM safety tests; accuracy checking
INTEG	Integration Agent	Cross-module consistency; broadcaster road map synthesis; data reconciliation
ANALYST	Pattern Analyst	Digital future trend analysis; competitive landscape; 5-year outlook
CAPCOM	HITL Interface	Human approval at Wave 1 and Wave 2 boundaries; only H-M channel
BUDGET	Resource Manager	Token budget tracking; wave completion reporting; model allocation monitoring
SURGEON	Health Monitor	Research quality degradation detection; factual consistency monitoring
TOPO	Topology Manager	Agent team structure; mid-mission restructuring if needed
ARCH	Architecture Agent	Cross-cutting structural decisions; document template enforcement
LOG	Chronicle Agent	Commit messages; milestone summaries; audit trail

5 Execution Summary

Metric	Value
Activation Profile	Fleet (6 modules, 17 roles)
Total Estimated Tokens	246,000
Model Split	Opus 24% / Sonnet 72% / Haiku 4%
Wave Count	4 waves
Parallel Efficiency	2 simultaneous tracks in Wave 1 (72% efficiency vs. sequential)
Estimated Sessions	4–6 Claude Code sessions
Estimated Context Windows	18–24
Human Gate Points	3 (Wave 1 start, Wave 2 start, final delivery)
Safety-Critical Tests	7 (all mandatory-pass / BLOCK)
Total Tests	45
Primary Regulatory Authority	FCC / 47 CFR Part 73
Research Domain	Broadcast history, regulatory law, spectrum engineering
Geographic Focus	Washington, Oregon, Idaho, California (West Coast primary)
Temporal Scope	1909–2026

> *"Before the internet, the radio dial was the only place where a single voice could reach the entire Pacific Northwest at once—in the dark, in the mountains, across the water. The frequency was free. The license was granted by the public. The obligation was to serve. That architecture still exists. The builder who learns to navigate it inherits something older and stranger than any streaming platform: the right to speak to a whole geography, and the responsibility to make that speech worth hearing."*
>
> — GSD Research Mission Vision, 2026 | Fox Infrastructure Group